GIBIConnect

Project Documentation

Ethiopia's verified higher education intelligence platform and grounded AI academic advisor

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Contents

1. Overview
2. Features
3. Technology Stack
4. Frontend Architecture
5. Backend Architecture
6. Database Design
7. Grounded AI Architecture
8. Deployment and Scalability
9. Project Team
10. Frequently Asked Questions

1. Overview

GIBIConnect is a web platform that unifies information about Ethiopian higher education in one place. It brings together **19 accredited universities**, **114 degree curricula**, **39 open-access curriculum documents**, verified admission cut-offs, statutory cost-sharing information, and an auditable AI advisor that answers only from verified data.

1.1 The problem

Every year, over 150,000 Ethiopian high school students sit for the Ethiopian University Entrance Examination (EUEE). When results are released, students, parents, and counselors face fragmented information:

- Cut-off points are scattered across informal Telegram channels.
- Statutory tuition rules under Proclamation No. 650/2009 are widely misunderstood.
- There is no single verifiable source for which departments, labs, and degree programs exist at institutions such as Addis Ababa University, Jimma, Bahir Dar, Dilla, or Wolkite.

This leads to misinformed university choices, mismatched department assignments, and high freshman attrition.

1.2 The solution

GIBIConnect provides a single, accessible gateway offering:

- A searchable directory of universities, campuses, faculties, departments, and programs.
- An EUEE cut-off and admissions hub with gender-disaggregated cut-offs and regional affirmative action.
- A statutory cost-sharing and financing calculator.
- A digital resource vault and a peer tutoring marketplace.
- A Retrieval-Augmented Generation (RAG) AI advisor with a deterministic fallback, guaranteeing continuous availability.

1.3 Key figures

Metric	Value
Verified institutions	19
Degree programs	114
Open-access curriculum documents	39
Upfront tuition for regular government-placed students	0 ETB
Post-graduation repayment cap	10% of gross monthly payroll

2. Features

2.1 Public landing page (`#home`)

The landing page uses an "Academic Deep Ocean" visual style and shows live institutional statistics. A real-time fuzzy search bar lets students search across universities, faculties, and degree fields.

2.2 Institutions Explorer (`#platform/institutions`)

- Founded year, governance status (for example, AAU's autonomous transition under Regulation No. 534/2023), and active leadership.
- Multi-campus breakdowns, such as 5 Kilo (AAiT), 4 Kilo (Natural Sciences), and Tikur Anbessa (Health Sciences) for Addis Ababa University.
- Department rosters with verified department heads, enrollment figures, and degree levels from Bachelor to Doctorate.
- Regional highlights, such as Dilla University's UNESCO-recognized Gedeo indigenous agroforestry research centers, the Odaya Referral Hospital clinical pavilion, and its Institute of Technology.

2.3 Programs catalog (`#platform/programs`)

Programs are organized by stream: **Natural Sciences** (STEM, Health) and **Social Sciences** (Business, Law, Humanities). Each program card shows:

- Degree level (B.Sc., B.A., M.Sc., MD, LL.B).
- Curriculum duration (4, 5, or 6 years for clinical medicine).
- Official credit allocations and career specializations.

2.4 Admissions Hub (`#platform/admissions`)

- Historical gender-disaggregated cut-offs (male and female pass marks).
- Regional affirmative action accommodations of +15 to +20 points for emerging regions such as Afar, Somali, Gambella, and Benishangul-Gumuz.
- Special needs allocations.
- An EUEE calculator where students enter matriculation scores.

2.5 Statutory cost-sharing and financing calculator

Public higher education in Ethiopia operates under statutory cost-sharing (Proclamation No. 650/2009). The calculator computes tuition liabilities:

- **0 ETB upfront** for regular government-placed students.
- Dining and dorm expenses are deferred until graduation.
- Repayment is a maximum **10% gross monthly payroll deduction**, beginning one year after employment.

2.6 Resource Vault (`#platform/resources`) and Peer Marketplace (`#platform/market`)

- **Resource Vault**: verified university lecture notes, past exit exam models, and national syllabi.
- **Peer Marketplace**: student tutors from institutions such as AAiT and ASTU offer 1-on-1 sessions in freshman mathematics, programming, and physics.

2.7 Grounded AI Academic Advisor (`#platform/ai-advisor`)

See Section 7 for full details.

2.8 Additional pages

- **About**: Ministry of Education and ETA accreditation context and platform mission.
- **Contact**: directory feedback and admissions inquiries.

- **Events**: national exit exams and the academic calendar.
- **Search Results**: real-time search filter interface.
- **Auth**: email login and Sign in with Google, with a profile manager and saved universities.

3. Technology Stack

Layer	Technology
Frontend	React 19, TypeScript 5.6+ (strict), Vite 8.3, Tailwind CSS v4, KaTeX 0.16.x, lucide-react
Routing	Zero-dependency hash-based router
Backend	Node.js, Express 4.x, strict TypeScript, layered service-repository pattern
Validation and security	Zod, JWT (12-hour expiry), bcrypt, RFC-7807 error responses
Database	PostgreSQL 16 on Neon Serverless, Prisma ORM, `pgvector`
AI	Google Gemini 2.0 / 3.5 Flash, `text-embedding-004` (768 dimensions), hybrid vector plus BM25 retrieval
Authentication	JWT with Google Identity Services (Sign in with Google)
Hosting	Vercel Edge CDN (frontend), Render containers (backend), Neon (database)

4. Frontend Architecture

4.1 Build tools and libraries

- **Language and runtime**: React 19 with TypeScript 5.6+ and strict type checking (`tsc -b`).
- **Build engine**: Vite 8.3 using the Rollup/Rolldown bundler with minification, tree-shaking, and code-splitting, with builds under 2 seconds.
- **Styling**: Tailwind CSS v4 using CSS variables and `@theme` utilities.
- **Typography**: serif stack for headings and Inter / system-ui sans-serif for body text.
- **Mathematics**: KaTeX 0.16.x for real-time LaTeX rendering.
- **Icons**: `lucide-react` for scalable SVG icons.

4.2 Layout and responsiveness

The interface avoids arbitrary fixed positioning and relies on responsive layouts. Platform grid:

```
grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 gap-6
```

Breakpoint	Layout
Mobile (below 640px)	Single-column stack with full-width cards and touch padding
Tablet (640px to 1024px)	Balanced 2-column grid
Desktop (above 1024px)	3-column layout with hover elevation (`hover:-translate-y-1 hover:shadow-lg`)

Campus detail layout: a 12-column master grid (`lg:grid-cols-12 gap-8`). The left column (`lg:col-span-8`) holds academic history, faculties, deans, and departments. The right column (`lg:col-span-4`) is a sticky admissions sidebar with cut-off cards and cost-sharing calculations.

AI Advisor workspace: a flexbox dual-pane layout. The left pane is a collapsible drawer (`w-72` to `w-0`) housing pinned chats and history. The right pane is the chat thread (`flex-1 flex flex-col`) with a sticky bottom scroll anchor (`messagesEndRef`).

4.3 Routing and state

- **Hash-based router**: parses and dispatches window hash changes with no runtime dependency. Routes: `#home`, `#about`, `#services`, `#contact`, `#auth`, `#platform/institutions`, `#campus/:id`, `#platform/programs`, `#platform/admissions`, `#platform/resources`, `#platform/market`, `#platform/ai-advisor`.
- **History integration**: listens to `hashchange` and `popstate` events to keep back and forward navigation working.
- **Session state**: `localStorage` holds the auth token (`gibiconnect_token`), the user profile (`gibiconnect_user`), saved universities (`gibiconnect_saved_institutions`), and AI chat histories.

4.4 Component map

Component / page	Responsibility
`App.tsx`	Hash router and application controller, platform header
`HomePage.tsx`	Hero, value proposition, highlights
`DirectoryGrid.tsx`	Responsive 3-column university card grid
`CampusDetailPage.tsx`	Deep institutional profile
`ProgramsPage.tsx`	Stream and level filters
`AdmissionsPage.tsx`	EUEE cut-off engine and quota analysis
`ResourcesPage.tsx`	Document vault
`MarketplacePage.tsx`	Peer tutoring
`AIAdvisorPage.tsx`	Dual-engine advisor and STEM tutor
`AuthPage.tsx`	Google Identity Services and JWT credentials
`UserProfileDropdown.tsx`	Saved universities, shortcuts, sign out

4.5 Directory structure

```
frontend/
├── public/
│   ├── favicon.ico, favicon.png
│   └── images/
│       ├── brand/
│       │   ├── gibiconnect-icon.png    # Official square emblem (G + Cap + Book)
│       │   └── gibiconnect-logo.png    # Full brand horizontal wordmark
│       └── logos/                      # Verified logos for all 19 universities
└── src/
    ├── App.tsx                         # Master router & platform header
    ├── index.css                       # Global Tailwind CSS & typography
    ├── main.tsx                        # React 19 entry point
    ├── components/
    │   ├── ai/InteractiveQuizRunner.tsx    # Interactive STEM assessment runner
    │   ├── directory/
    │   │   ├── DirectoryGrid.tsx           # 3-column university cards grid
    │   │   └── InstitutionCard.tsx         # Campus card with cutoff badge
    │   └── layout/
    │       ├── Navbar.tsx                  # Public header with view routing
    │       └── UserProfileDropdown.tsx     # Avatar menu with bookmarks
    ├── config/api.ts                   # Render / Vercel API URL resolver
    ├── data/
    │   ├── mockInstitutions.ts         # Registry of 19 Ethiopian universities
    │   └── universityDetails.ts        # Faculties, departments, deans & rosters
    ├── pages/                          # About, Admissions, AIAdvisor, Auth,
    │                                   # CampusDetail, Contact, Events, Home,
    │                                   # Marketplace, Programs, Resources,
    │                                   # SearchResults
    ├── types/
    │   ├── admission.ts                # Cutoff & quota interfaces
    │   └── institution.ts              # Institutional entity schemas
    └── utils/
        ├── clientSvgGenerator.ts       # Generative SVG diagrams for STEM concepts
        └── resolveInstitution.ts       # Typo-tolerant university entity resolver
```

5. Backend Architecture

5.1 Layered design

Every feature module follows a four-layer separation of concerns:

1. **Routes** (`*.routes.ts`): define endpoints and HTTP verbs, and mount authentication and validation middleware.
2. **Controllers** (`*.controller.ts`): parse request parameters, enforce Zod schemas, set HTTP status codes, and format JSON responses.
3. **Services** (`*.service.ts`): hold business logic, calculations, multi-table operations, and external API calls.
4. **Repositories** (`*.repository.ts`): run SQL through Prisma and the native `pg.Pool` connection layer.

5.2 API surface

Route prefix	Module purpose
`/api/v1/auth`	Registration, bcrypt hashing, JWT issuance (12-hour expiry), Google OAuth (`POST /api/v1/auth/google`)
`/api/v1/institutions`	Multi-campus queries, regional filters, faculty rosters
`/api/v1/programs`	Curriculum search, ECTS credit mapping, degree duration lookups
`/api/v1/admissions`	EUEE criteria, historical cut-offs, affirmative action algorithms
`/api/v1/ai`	Academic advising, STEM tutoring, document Q&A

The `finance` module implements statutory cost-sharing and tuition calculations under Proclamation No. 650/2009. Further modules cover `events`, `marketplace`, `resources`, and `users`.

5.3 Middleware and security

- `auth.middleware.ts`: validates `Bearer <token>` against `JWT_ACCESS_SECRET` and attaches `req.user` with `sub`, `email`, and `role`.
- `error.middleware.ts`: normalizes Zod failures, Prisma constraint errors, and unexpected exceptions into RFC-7807 error responses.
- `fix-esm-imports.mjs`: a post-compilation script that adds explicit `.js` extensions to emitted files for native Node.js ESM execution.

5.4 Directory structure

```
backend/
├── prisma/
│   ├── schema.prisma                  # Relational model & pgvector definitions
│   ├── seed.ts                        # Seeds the 19 Ethiopian universities
│   └── neon-dump.sql                  # Canonical Neon PostgreSQL migration dump
└── src/
    ├── app.ts                         # Express setup & middleware assembly
    ├── server.ts                      # Server entry point (port 5000)
    ├── fix-esm-imports.mjs            # Post-build ESM import patcher
    ├── config/
    │   ├── database.ts                # pg.Pool connection pooler
    │   └── env.ts                     # Zod-validated environment configuration
    ├── middleware/
    │   ├── auth.middleware.ts         # JWT authentication & role enforcement
    │   └── error.middleware.ts        # Centralized exception formatter
    └── modules/
        ├── admissions/                # EUEE cutoffs & quota services
        ├── ai/
        │   ├── ai.advisor.service.ts  # Multi-stage grounding & prompt orchestration
        │   ├── ai.controller.ts       # AI chat & document upload endpoints
        │   ├── ai.routes.ts           # /api/v1/ai routes
        │   ├── ai.types.ts            # Query & intent type definitions
        │   ├── document.chunker.ts    # Recursive semantic text chunking
        │   ├── document.parser.ts     # PDF & plain-text curriculum extractor
        │   ├── image.generator.service.ts  # Visual concepts generator
        │   ├── llm.client.ts          # Resilient Google Gemini client
        │   ├── rag.corpus.ts          # Verified higher education corpus
        │   └── rag.retriever.ts       # Hybrid vector/keyword retrieval engine
        ├── auth/                      # Authentication & Google GIS handler
        ├── events/                    # Academic calendar & national exit exams
        ├── finance/                   # Cost-sharing & credit fees
        ├── institutions/              # Campus and faculty directories
        ├── marketplace/               # Peer tutor exchange
        ├── programs/                  # Degree catalog
        ├── resources/                 # Educational document vault
        └── users/                     # Student profile management
```

6. Database Design

The database runs on **PostgreSQL 16** hosted on **Neon Serverless**, with connection pooling and the `pgvector` extension.

6.1 Entity hierarchy

The core academic hierarchy is: **Institutions → Campuses → Faculties → Departments → Programs**. Programs in turn own **Admission Cut-offs** and **Tuition Rates**.

- Users write reviews, bookmark institutions, own AI conversations, and list marketplace items.
- AI conversations contain AI messages.
- Institutions publish resources.
- Document chunks reference institutions.

6.2 Table specifications

Table	Primary key	Key foreign keys	Purpose
`users`	`id` (UUID)	None	Profiles, hashed passwords, roles (`admin`, `student`, `tutor`), Google account bindings
`institutions`	`id` (UUID)	None	All 19 universities: official codes (`AAU`, `ASTU`, `DU`), Amharic/English names, websites, logos
`campuses`	`id` (UUID)	`institution_id` → `institutions.id`	Physical campuses (AAiT 5 Kilo, Odaya Campus Dilla, Tibebe Ghion Bahir Dar)
`faculties`	`id` (UUID)	`campus_id` → `campuses.id`	Colleges, schools, and institutes with deans and established years
`departments`	`id` (UUID)	`faculty_id` → `faculties.id`	Departments with heads, enrollment, and specializations
`programs`	`id` (UUID)	`department_id` → `departments.id`	Accredited degrees, duration, ECTS credits, stream (`NATURAL` / `SOCIAL`)
`admission_cutoffs`	`id` (UUID)	`program_id` → `programs.id`	Historical pass marks by gender and region
`tuition_rates`	`id` (UUID)	`program_id` → `programs.id`	Statutory cost-sharing flags and extension per-credit-hour costs in ETB
`resources`	`id` (UUID)	`institution_id` → `institutions.id`	Syllabus outlines, past national exams, lecture notes
`document_chunks`	`id` (UUID)	`institution_id` → `institutions.id`	768-dimension vector embeddings for RAG semantic search

7. Grounded AI Architecture

7.1 Request flow

1. The student submits a question on `AIAdvisorPage.tsx`, for example "What departments are in Dilla and Jimma?"
2. `resolveInstitution.ts` normalizes typos ("dila" becomes "dilla") and extracts the institutions (`DU`, `JU`).
3. The page sends `POST /api/v1/ai/chat` with the message and session ID.
4. If the backend is available, `rag.retriever.ts` runs a hybrid (vector plus keyword) retrieval for each extracted university.
5. The system prompt, retrieved chunks, and user message go to Google Gemini.
6. If Gemini responds, the grounded multi-university answer is returned. If Gemini hits latency or rate limits, the Grounded Rule Synthesizer builds a deterministic, verified department directory instead.
7. If the network is offline, the client-side grounded knowledge engine produces the answer instantly.
8. The page renders structured academic tables and Markdown.

7.2 Dual-inference architecture

- **Primary model**: Google Gemini 2.0 / 3.5 Flash via REST, wrapped in a custom Node.js HTTPS agent. On local Windows environments the agent sets `rejectUnauthorized: false` to bypass system CA inspection issues.
- **Secondary synthesizer**: an internal deterministic synthesizer that builds verified responses from the Ethiopian Ministry of Education catalog and Proclamation No. 650/2009 rules whenever external APIs are unavailable.

7.3 Typo-tolerant multi-university entity extraction

- **Phonetic and typo normalization**: `dila` → `dilla`, `unviesrity` / `unvisridty` → `university`, `welkite` → `wolkite`, `gonder` → `gondar`, `meqele` → `mekelle`.
- **Multi-university extraction**: "Compare software engineering in AAU and ASTU" returns `['AAU', 'ASTU']` with a separate section for each.
- **Exhaustive aliases**: mappings for all 19 universities, including campus names (5 Kilo, 4 Kilo, Odaya, Gubre, Tibebe Ghion, Hiwot Fana, Tikur Anbessa).

7.4 RAG pipeline and semantic chunking

- **Chunking**: `document.chunker.ts` splits documents on semantic boundaries into 500 to 800 token chunks with a 100-token overlap.
- **Embeddings**: Google `text-embedding-004` produces 768-dimensional vectors, matched by cosine distance.
- **Hybrid search**: dense vector similarity is combined with BM25 keyword matching so that exact institutional codes (`AASTU`, `ASTU`, `AMU`, `DMU`, `DDU`) are never conflated.

7.5 Interactive 6-stage STEM tutor

For freshman STEM courses such as Calculus (MATH1011) and Physics (PHYS1011), the tutor guides students through six stages:

1. **Concept introduction**: key principles (Newton's 2nd Law, 2D projectile motion, derivatives, chemical equilibrium).
2. **Mathematical formulation**: equations in LaTeX rendered with KaTeX.
3. **Worked examples**: typical Ethiopian freshman exam problems solved step by step.
4. **Interactive practice problem**: a generated challenge question.
5. **Real-time evaluation**: analyzes the answer and pinpoints algebra or physics errors.
6. **Concept mastery badge**: updates progress and unlocks advanced topics.

Example formulation for 2D projectile motion (vertical position over time):

$$y(t) = y_0 + v_{0y} \cdot t - \tfrac{1}{2} \cdot g \cdot t^2$$

7.6 Generative visuals

When a student asks for a diagram (for example, a heart cross-section or an electrical power distribution grid), the system generates an interactive schematic rendered directly as SVG.

8. Deployment and Scalability

- **Frontend**: statically generated with Vite and cached on the Vercel Edge CDN.

- **Backend**: containerized and hosted on Render, listening on port 5000.
- **Database**: Neon Serverless PostgreSQL, which scales compute automatically with demand.
- **Low-bandwidth accessibility**: no heavy UI frameworks, sub-200ms load time, an offline client fallback for advising, and responsive grids that adapt to budget smartphones.

9. Project Team

GIBIConnect was developed by the following team members:

No.	Name	ID
1	Fekadu Alemnew	CTC-4438-26
2	Ermiyas Cheru	CTC/5723/26

10. Frequently Asked Questions

Question	Answer
Where does the university and cut-off data come from?	Records are mapped to official Ministry of Education registrar bulletins, verified EUEE matrices, and the statutory cost-sharing framework under Proclamation No. 650/2009.
How does the AI avoid hallucinating?	It uses RAG with a dual-inference pipeline and answers only from verified institutional chunks. If confidence is low, the deterministic synthesizer returns verified registry data.
How does the system handle high traffic during EUEE result releases?	The frontend is static and cached on the Vercel Edge CDN, the backend is containerized on Render, and Neon scales database compute automatically.
Can students on slow internet or mobile devices use it?	Yes. The platform avoids heavy UI frameworks, loads in under 200ms, supports an offline client fallback for advising, and uses responsive grids for budget smartphones.